

Basic properties of numerical inequalities

What you may do to both sides — and the one operation that turns the sign around.

LESSON 40–41

2 × 40 MIN

ALGEBRA 8, §12

STAGE 9 · 4.3

LESSON CLOCK

5'

12'

8'

13'

2'

Warm-up	5 min
Explanation	12 min
Interactive	8 min
Practice	13 min
Homework	2 min

LEARNING OBJECTIVES

- State the four basic properties of inequalities.
- Multiply or divide by a negative number and reverse the sign.
- Use transitivity to chain comparisons.
- Decide which operations preserve an inequality and which do not.

TEXTBOOK

National	§12, pp. 71–74
Cambridge	Learner’s Book pp. 96–102
Workbook	Workbook 4.3

01

Explanation

The four properties

FOR ANY NUMBERS a , b , c

- Transitivity.** If $a > b$ and $b > c$ then $a > c$.
- Adding.** If $a > b$ then $a + c > b + c$ — for **any** c , positive or negative.
- Multiplying by a positive number.** If $a > b$ and $c > 0$ then $ac > bc$.
- Multiplying by a negative number.** If $a > b$ and $c < 0$ then $ac < bc$ — **the sign reverses**.

Property 2 is why you may move a term from one side to the other, changing its sign, exactly as in an equation. Property 4 is the one thing an inequality does that an equation does not.

Why the sign reverses

Take $3 > 2$, which is certainly true, and multiply both sides by -1 :

-3 and -2

On the number line -3 sits to the **left** of -2 , so $-3 < -2$. The order has turned round.

Multiplying by a negative number reflects both numbers through 0, and reflection swaps left and right.

THE RULE THAT COSTS THE MOST MARKS ALL YEAR

From $-2x > 6$ you get $x < -3$, not $x > -3$. Dividing by -2 reverses the sign. Write the new sign **first**, then do the arithmetic.

Two consequences worth knowing

- **Reciprocals.** If $a > b > 0$ then $\frac{1}{a} < \frac{1}{b}$. Bigger denominator, smaller fraction.
- **Squares.** If $a > b > 0$ then $a^2 > b^2$. But this needs both numbers positive: $-5 > -7$ while $25 < 49$.

A QUICK TEST TO BUILD THE HABIT

Before applying any step, ask: "am I multiplying or dividing by something negative?" If yes, the sign turns. If no, it stays.

02 Worked examples

EXAMPLE 1 Given $a > b$, compare $3a - 5$ and $3b - 5$

① $a > b$

Given.

② $3a > 3b$

Multiply by 3, which is positive — sign unchanged.

③ $3a - 5 > 3b - 5$

Subtract 5 from both sides — sign unchanged.

ANSWER

$$3a - 5 > 3b - 5$$

EXAMPLE 2 Given $a > b$, compare $-2a$ and $-2b$

① $a > b$

Given.

② Multiplying by -2 , a negative number.

Property 4 applies.

③ $-2a < -2b$

The sign reverses.

ANSWER

$$-2a < -2b$$

EXAMPLE 3 Given $2 < x < 5$, find the range of $3 - 2x$

① $2 < x < 5$
Given.

② $-4 > -2x > -10$
Multiply throughout by -2 — both signs reverse.

③ $-10 < -2x < -4$
Rewrite the smaller number on the left.

④ $-7 < 3 - 2x < -1$
Add 3 throughout — signs unchanged.

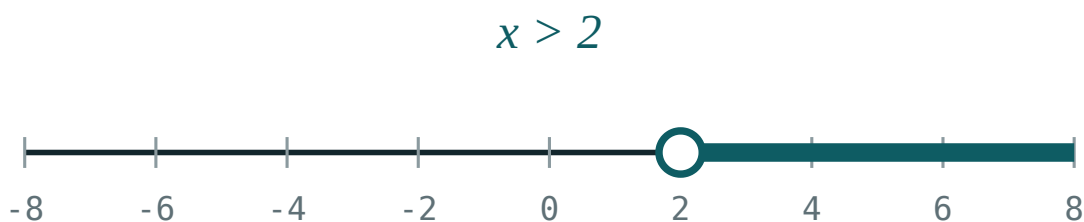
ANSWER

$$-7 < 3 - 2x < -1$$

03 Interactive model

Use the flip button to show that the same boundary gives two different solution sets.

Which side, and is the end included?



inequality $x > 2$

boundary **2 (open)**

$x = 0$ works? **no**

$x = 6$ works? **yes**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Property of an inequality	Tengsizlik xossasi	Свойство неравенства
Transitivity	Tranzitivlik	Транзитивность
Reverse the sign	Ishorani almashtirish	Изменить знак на противоположный
Both sides	Ikkala tomon	Обе части
Multiply by	Ko'paytirish	Умножить на
Divide by	Bo'lish	Разделить на
Preserve	Saqlanadi	Сохраняется
Equivalent inequality	Teng kuchli tengsizlik	Равносильное неравенство

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

From $a > b$ it follows that:

From $a > b$ and $c < 0$ it follows that:

$-3x > 12$ gives:

If $a > b > 0$ then:

$$\frac{1}{a} > \frac{1}{b}$$

$$\frac{1}{a} < \frac{1}{b}$$

$$\frac{1}{a} = \frac{1}{b}$$

it depends

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $a > b$. Compare $a + 4$ and $b + 4$

ANSWER $a + 4 > b + 4$

2. $a > b$. Compare $a - 7$ and $b - 7$

ANSWER $a - 7 > b - 7$

3. $a > b$. Compare $5a$ and $5b$

ANSWER $5a > 5b$

4. $a > b$. Compare $-a$ and $-b$

ANSWER $-a < -b$

5. Solve $-x > 3$

ANSWER $x < -3$

6. Solve $2x > 8$

ANSWER $x > 4$

7. Solve $-2x \geq 10$

ANSWER $x \leq -5$

Medium · the standard question

7 PROBLEMS

1. $a > b$. Compare $3a - 5$ and $3b - 5$

ANSWER $3a - 5 > 3b - 5$

2. $a > b$. Compare $-2a + 1$ and $-2b + 1$

ANSWER $-2a + 1 < -2b + 1$

3. $a > b > 0$. Compare $\frac{1}{a}$ and $\frac{1}{b}$

ANSWER $\frac{1}{a} < \frac{1}{b}$

4. $a > b > 0$. Compare a^2 and b^2

ANSWER $a^2 > b^2$

5. Solve $-4x + 3 < 11$

ANSWER $x > -2$

6. $2 < x < 5$. Find the range of $2x + 1$

ANSWER $5 < 2x + 1 < 11$

7. Solve $\frac{x}{-3} > 2$

ANSWER $x < -6$

Hard · stretch and reason

7 PROBLEMS

1. $2 < x < 5$. Find the range of $3 - 2x$

ANSWER $-7 < 3 - 2x < -1$

2. $-1 < x < 4$. Find the range of $5 - x$

ANSWER $1 < 5 - x < 6$

3. $a > b$. Is it always true that $a^2 > b^2$? Give a counter-example.

ANSWER No — $-5 > -7$ but $25 < 49$. It needs $a > b > 0$.

4. $a > b$ and $c > d$. Prove $a + c > b + d$

ANSWER $a + c > b + c$ and $b + c > b + d$, then transitivity.

5. $3 < x < 7$ and $1 < y < 2$. Find the range of $x - y$

ANSWER $1 < x - y < 6$

6. $a > b > 0$. Compare $\frac{a}{b}$ and 1

ANSWER $\frac{a}{b} > 1$ — divide $a > b$ by the positive number b .

7. Solve $5 - 3x \leq 2x - 10$

ANSWER $15 \leq 5x$, so $x \geq 3$

07

Homework

Homework — 6 problems

Algebra 8, §12, pp. 71–74. Say out loud whether the sign turns before you write each line.

- $a > b$. Compare $4a + 3$ and $4b + 3$
- $a > b$. Compare $-5a$ and $-5b$
- Solve $-3x + 2 > 14$
- $1 < x < 6$. Find the range of $4 - x$

5. $a > b > 0$. Compare $\frac{2}{a}$ and $\frac{2}{b}$

6. Give a counter-example to “ $a > b$ implies $a^2 > b^2$ ”